Keaton Van Patten

AI Systems Engineer

Rancho Cordova, CA · remote or Sacramento area · keatonvp123@gmail.com · github.com/Peacho97 · peacho97.github.io

SUMMARY

Builder-operator who ships production AI systems end to end and measures them honestly. Sole developer of a 200K-line agentic platform (678 API endpoints, 122-tool agent, ~3,000 tests) and of a live scheduling system serving a healthcare clinic. Every architecture decision traces to a number measured on real hardware — including the ones that proved my own claims wrong.

SELECTED PRODUCTION SYSTEMS

Clinic Scheduling System + Outage Failover — *pediatric therapy clinic (live since Aug 2026)*

- Sole developer of a production workload-based scheduling system replacing a fully paper process: ~35K lines, 30 REST endpoints, 15 role-gated pages (PHP 8.3 / SQLite / zero-dependency JS PWA); in daily use by clinic staff (573 audited actions in first 18 days).
- Designed an 8-component weekly workload model computing therapist utilization to the quarter-hour with hard over-capacity blocking, plus a filter-and-rank substitute selector for absence coverage.
- After discovering a real production outage, designed, built, and deployed a zero-dependency Node failover mailer on Fly.io in under 48 hours — outbound-only (no public IP), server-side PHI minimization so only de-identified case codes ever leave the host; 157 tests including both DST transitions.
- Commissioned a security audit of my own application pre-launch via my own security-review desk; remediated two critical findings (install-takeover path, cross-role data exposure); shipped Argon2id auth, email OTP 2FA, CSRF hardening, and a 14-action audit log. Wrote 800+ integration test assertions with no third-party test framework.

RAG Reference Service ("Librarian") — *production retrieval over a 19K-document corpus*

- Built and operate a retrieval-augmented Q&A service over 3,190 files / 14,534 code symbols / 18,094 embedding vectors: hybrid Okapi BM25 + dense-vector retrieval fused with Reciprocal Rank Fusion, custom reranking, incremental content-hash embedding (21.8× measured batch speedup).
- Served to AI agents as an 8-tool MCP server — 613 live queries in 30 days, 67% machine-to-machine — through a 5-provider LLM fallback chain (~92% answered by the low-cost primary).
- Built a CI-gated evaluation harness (22 graded questions, release blocked under 90%, deterministic questions must pass with every model provider offline); drove measured accuracy from 86.4% to 100%.

Autonomous Incident Desk — *three-tier deflection ladder*

- Built a self-healing service desk monitoring 15 production backends on a 15-second probe loop: 612 incidents in 53 days, 72.9% auto-resolved at a 55-second mean time-to-resolution, 0.33% escalated to a human.
- Architected cost-tiered deflection — deterministic playbooks first (75.6% first-line success), LLM triage only when they fail (3.3% of tickets), diagnosis-only model contract, circuit breakers and a kill-pattern validator bounding blast radius.
- Scope: these are infrastructure incidents, not customer-support tickets — the transferable part is the deflection architecture, not the domain.

Multi-Provider LLM Platform — *cost governance and orchestration*

- Operate a metered fleet spanning 21 models across 8 providers (Anthropic, DeepSeek, local Ollama, more) under a USD-exact per-call ledger: 2,129 calls, 41M tokens, 38 attributed use-case surfaces, 64% prompt-cache hit rate.
- Cut cost per model call 71% while volume grew 51% by profiling provider pricing, latency, and data-sensitivity, then re-routing background workloads to cheaper tiers and pinning latency-critical and private-data paths to local GPUs.
- Stood up a self-hosted LiteLLM control plane (PostgreSQL-backed) issuing per-use-case virtual API keys with hard monthly budgets, with a raw-socket reachability probe driving automatic gateway→direct fallback (852 calls served uninterrupted through the fallback path).

Personal AI Platform ("Companion Hub") — *122-tool production agent*

- Built a local-first agentic platform (Python/FastAPI + Svelte/Electron; 678 HTTP endpoints, ~3,000 tests) whose agent dispatches 122 real tools — filesystem, shell, calendar, health, messaging — under a three-layer authorization model (fail-closed per-surface allowlists, chokepoint denylists, credential-aware filesystem perimeter; 32 dedicated gate tests).
- Designed a token-budgeted 5-zone prompt architecture holding the static prefix byte-identical across turns for inference-cache hits; cut small-model tool-catalog overhead 5.3× (7.4K → 1.4K tokens) with a curated subset plus scored tool search.
- Built an AI security-review desk (110 consultations, 165 specialist reviews, 548 tracked checklist items) with adversarially-benchmarked local triage — the selected model resisted 6/6 prompt injections; a candidate was rejected for obeying injected verdicts.
- Run dual-GPU local inference infrastructure: a cross-process GPU lease broker (priority aging, fencing tokens, 73-second measured failover), video generation on AMD ROCm at 87% VRAM utilization, and a 744B-parameter MoE model tuned +86% on consumer hardware.

EXPERIENCE

HRBC Inc. — Systems Manager & Software Developer · Sacramento, CA Sept 2020 – Present

- Sole technical owner for a mortgage brokerage: systems administration, Microsoft 365 / Entra ID, user access management, vendor systems (LOS, CRM, pricing engines), incident response, and training for loan officers — 5 years of direct requirements discovery with non-technical stakeholders.
- Built an internal operations hub with an embedded AI assistant: hand-rolled agentic tool-use loop on the Anthropic API (8 typed JSON-schema tools, bounded round-trips, mutation tracking) over a 15-system vendor registry; security policy encoded in the prompt and tool schemas — credential locations stored, never secrets. The agentic backend is complete and demonstrated via API; the UI was never wired up.
- Originated loans and advised clients directly; trained and supervised loan officers and assistants.

MP NexLevel — Utility Technician 2018 – 2020

Crew lead on utility infrastructure projects; equipment operation and maintenance.

SKILLS

AI/LLM: agent orchestration · multi-agent systems · RAG (BM25, vector embeddings, RRF hybrid search) · tool calling / function calling · MCP (Model Context Protocol) · prompt engineering · LLM evaluation harnesses · LLM-as-judge · guardrails & prompt-injection defense · human-in-the-loop gating · cost attribution & model routing · local inference (Ollama, llama.cpp, CUDA, ROCm) · Claude API · OpenAI-compatible APIs · DeepSeek · Whisper STT · TTS

Engineering: Python · FastAPI · TypeScript/JavaScript · SQL/SQLite/PostgreSQL · Svelte · Electron · REST API design · WebSockets/SSE · PHP · Docker · CI/CD (GitHub Actions) · pytest · Fly.io · Tailscale · Windows/Linux administration

Customer-facing: requirements discovery · stakeholder management · technical training · incident communication · security-review facilitation

EDUCATION

Google IT Support Certification · Google Computer Networking Certification